Here are some questions that you should be able to answer comfortably. Please do them to see whether you are. If you have difficulties, you have forgotten your algebra and elementary calculus. Please take remedial action.

1. For $x \in \mathbb{R}$, compute

   $1 + x + x^2 + \cdots + x^7 =$

2. For $0 < x < 1$, compute

   (a) $1 + x + x^2 + \cdots =$

   (b) $1 + 2x + 3x^2 + \cdots =$

   (c) $x + \frac{x^2}{2} + \frac{x^3}{3} + \cdots =$

3. For $x \in \mathbb{R}$, state the value of

   $1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \cdots =$

4. It is known that, for every $\lambda > 0$ and every integer $k \geq 0$,

$$\int_0^\infty \frac{\lambda e^{-\lambda x}(\lambda x)^k}{k!}\, dx = 1.$$

   Using this fact, compute

   (a) $\int_0^\infty e^{-2x} x^5 \, dx =$

   (b) $\int_0^\infty x^2 \frac{e^{-x} x^5}{5!} \, dx =$